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(57) Abstract:

An augmented reality based intelligent assembly guidance and torque monitoring system (100) for industrial valve manufacturing is disclosed. The system (100) comprises an AR head-mounted display (HMD) subsystem (120) with an integrated forward-facing camera (121), a CNN-based vision subsystem (130) trained on labelled images of valve sub-components, an assembly guidance and rendering engine (140) implementing a finite state machine (141), a torque monitoring subsystem (150) comprising torque sensors (151) coupled to assembly tools (152) and a torque alert engine (153), a proximity sensor subsystem (160) comprising an ultrasonic sensor (161), and an assembly traceability logging module (170). The CNN module (130) performs real-time part classification and orientation verification; state transitions are gated by CNN confidence scores of 0.85 or greater, enforcing Poka-Yoke error-proofing at each step. The torque alert engine (153) compares real-time torque values against configurable thresholds per fastener pos

FORM 2
THE PATENTS ACT, 1970 (39 of 1970)
&
THE PATENTS RULES, 2003
COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION

**AUGMENTED REALITY BASED INTELLIGENT ASSEMBLY GUIDANCE AND
TORQUE MONITORING SYSTEM FOR INDUSTRIAL VALVE MANUFACTURING**

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3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD

The present disclosure relates to the field of industrial assembly automation and quality control technologies. More particularly, the present disclosure relates to an Augmented Reality (AR) based intelligent assembly guidance system for multi-component valve manufacturing that integrates Convolutional Neural Network (CNN) based real-time part recognition, step-by-step three-dimensional holographic assembly instructions rendered through an AR head-mounted display (HMD), and a sensor-integrated real-time torque monitoring module with threshold-based alert generation, and a method thereof.

DEFINITIONS

For the purpose of this specification, the following terms shall have the meanings assigned to them below. These definitions are provided to clarify terminology used throughout the description and claims and are not intended to limit the scope of the invention.

"Augmented Reality Head-Mounted Display (AR HMD)" refers to a binocular see-through wearable display device incorporating a forward-facing camera, stereo optical display panels with a minimum 60-degree horizontal field-of-view, an inertial measurement unit (IMU), and wireless transceiver, configured to overlay digital holographic content onto the wearer's physical workspace.

"Convolutional Neural Network (CNN) Module" refers to a deep learning inference engine trained on labelled images of valve sub-components, configured to perform real-time part classification, bounding box detection, orientation verification, and confidence score generation from camera frames captured by the AR HMD.

"Assembly Guidance Rendering Engine" refers to a software module that generates and streams three-dimensional holographic step-by-step assembly instructions into the operator's field-of-view via the AR HMD, with state transitions gated by CNN inference confirmation results.

"Torque Monitoring Subsystem" refers to an assembly of one or more torque sensors embedded in or wirelessly coupled to assembly tools, an alert engine, and a threshold configuration database, configured to measure real-time torque applied to fasteners and generate visual alerts within the AR display upon deviation from pre-defined minimum and maximum torque thresholds.

"Finite State Machine (FSM)" refers to the assembly sequence controller that encodes each valve assembly step as a discrete state, wherein transitions between states are triggered exclusively by CNN inference confidence scores exceeding a configurable threshold value.

"Ultrasonic Proximity Sensor" refers to an ultrasonic distance measurement device operating at 40 kHz, configured to confirm final assembly completion when the measured distance between the sensor and the completed assembly falls below a pre-defined threshold of 10 centimeters.

"Assembly Traceability Log" refers to a digital serialized record generated for each assembled valve unit, recording component identifiers, CNN confidence scores, torque measurements per joint, assembly timestamps, and operator identification credentials.

"Poka-Yoke" refers to error-proofing mechanisms implemented in the system that physically or electronically prevent an operator from proceeding to a subsequent assembly step without verified correct completion of the current step.

BACKGROUND

The background information herein below relates to the present disclosure but is not necessarily prior art.

Industrial assembly of multi-component valves including ball valves, gate valves, butterfly valves, and check valves requires precise sequential assembly of numerous sub-components such as seals, seats, fasteners, actuators, pins, rings, and end-caps. The assembly process relies predominantly on human operators following paper-based or static digital work instructions, resulting in assembly errors including missing components, incorrect part orientation, out-of-sequence assembly, and incorrect torque application that lead to product failures, safety hazards, and costly recalls.

Existing computer vision-based quality control systems operate as post-assembly inspection systems that identify defects only after the complete assembly is finished, resulting in scrap, rework costs, and production delays. There is a critical need for an in-process, real-time verification system that confirms correct assembly at each individual step before the operator proceeds.

Torque control during fastener tightening is a critical quality parameter in valve manufacturing. Under-torqued joints cause fluid leaks and mechanical failures; over-torqued joints cause thread stripping and component cracking. Existing digital torque tools operate in isolation without integration into assembly guidance or augmented reality visualization workflows.

Augmented Reality technology has demonstrated potential in industrial training and guidance; however, prior art systems lack the combination of CNN-based sequential part verification, AR-rendered 3D assembly overlays, and real-time torque sensor integration in a unified closed-loop industrial assembly guidance system. The present disclosure addresses all these deficiencies.

Given these factors, it is crucial to address the above-mentioned challenges, and therefore a need was felt for an augmented reality based intelligent assembly guidance and torque monitoring system for industrial valve manufacturing that eliminates assembly errors, enforces torque compliance, and enables full digital traceability of assembly operations.

OBJECTS

Some of the objects of the present disclosure, which at least one embodiment herein satisfies, are as follows:

It is an object of the present disclosure to ameliorate one or more problems of the prior art or to at least provide a useful alternative.

An object of the present disclosure is to provide an AR-based system that superimposes real-time, step-by-step three-dimensional holographic assembly guidance directly onto an operator's field-of-view using an AR head-mounted display device.

Another object of the present disclosure is to implement a CNN-based object recognition module trained to identify individual valve sub-components, verify their correct orientation, and confirm each assembly step before proceeding to the next.

Still another object of the present disclosure is to integrate real-time torque monitoring sensors with the AR guidance system such that torque applied to each fastener is continuously compared against pre-defined thresholds and alerts are rendered within the AR display upon threshold violation.

Yet another object of the present disclosure is to provide an ultrasonic proximity sensor-based final assembly verification mechanism that confirms completion of the full assembly sequence when the sensor measures a distance of 10 centimeters or less.

Still another object of the present disclosure is to generate a digital serialized assembly traceability log for each unit, recording component identifiers, CNN confidence scores, torque values per joint, timestamps, and operator credentials.

Yet another object of the present disclosure is to substantially reduce assembly errors, improve operator training efficiency, enforce Poka-Yoke error-proofing at each assembly step, and improve overall product quality in industrial multi-component valve manufacturing.

Other objects and advantages of the present disclosure will be more apparent from the following description, which is not intended to limit the scope of the present disclosure.

SUMMARY

The present disclosure envisages an augmented reality based intelligent assembly guidance and torque monitoring system for industrial valve manufacturing. The system comprises an AR head-mounted display (HMD) subsystem, a CNN-based vision subsystem, an assembly guidance and rendering engine, a torque monitoring subsystem, a proximity sensor subsystem, a central processing unit, and an assembly traceability logging module.

The AR HMD subsystem incorporates a forward-facing RGB camera, stereo display panels, an IMU, and a wireless transceiver configured to overlay holographic assembly guidance content onto the operator's physical workspace in real time.

The CNN-based vision subsystem employs a convolutional neural network architecture trained on a dataset of labelled images of valve sub-components. The CNN is configured to classify each component, detect its bounding box, verify orientation alignment, and return a confidence score for each camera frame.

The assembly guidance and rendering engine implements a finite state machine with discrete states corresponding to each assembly step from S0 (idle) through S5 (assembly complete). State transitions are gated by CNN confidence scores exceeding a configurable threshold of 0.85, preventing the operator from proceeding without verified completion of the current step.

The torque monitoring subsystem comprises one or more torque sensors communicatively coupled to assembly tools such as wrenches and nutrunners via Bluetooth 5.0 or wired RS-485 interface, and an alert engine that compares instantaneous torque values against configurable minimum and maximum thresholds per fastener position. A yellow caution overlay is rendered in the AR display when torque falls below the minimum threshold; a red danger alert with audible signal is generated when torque exceeds the maximum threshold.

The proximity sensor subsystem uses an ultrasonic sensor to confirm final assembly completion when the measured distance between the sensor and the completed assembly is 10 centimeters or less.

In an embodiment, the CNN inference module employs a YOLO or ResNet-50 backbone architecture trained on a dataset comprising no fewer than 5,000 annotated images per valve component class.

In an embodiment, the assembly guidance rendering engine streams 3D animation video clips for each assembly step into the AR HMD display, enabling the operator to follow visual demonstrations of each assembly action.

In an embodiment, the system further comprises a cloud-based analytics module configured to store assembly traceability logs, CNN performance data, torque records, and system usage patterns to support long-term quality improvement and regulatory compliance.

The present disclosure further envisages a method for AR-guided industrial assembly with torque monitoring. The method comprises the following steps:

- detecting and classifying individual valve sub-components using a trained CNN model receiving image frames from the AR HMD camera;
- displaying three-dimensional assembly guidance overlays within the operator's field-of-view via the AR HMD upon successful component detection and orientation verification;
- monitoring real-time torque applied to fasteners via integrated torque sensors during each tightening operation;
- comparing measured torque values against stored threshold ranges and generating AR-rendered visual and audible alerts upon violation;
- confirming final assembly completion via ultrasonic proximity sensor when the measured distance is 10 centimeters or less; and
- logging all assembly step data including component identifiers, CNN confidence scores, torque values, timestamps, and operator credentials for traceability records.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

An augmented reality based intelligent assembly guidance and torque monitoring system for industrial valve manufacturing, and a method thereof, will now be described with the help of the accompanying drawings, in which:

Figure 1 illustrates a complete system architecture block diagram of the augmented reality based intelligent assembly guidance and torque monitoring system, showing all subsystems and their interconnections, in accordance with an embodiment of the present disclosure;

Figure 2 illustrates the assembly sequence finite state machine (FSM) and CNN inference pipeline flow diagram showing the step-by-step assembly states, transition conditions, and error-proofing logic, in accordance with an embodiment of the present disclosure;

Figure 3 illustrates a schematic representation of the torque monitoring subsystem alert logic, the valve component identification drawing for all five sub-components (Parts 1 through 5), and the AR operator field-of-view schematic showing the holographic overlay and HUD elements, in accordance with an embodiment of the present disclosure.

LIST OF REFERENCE NUMERALS

- 100 -- Augmented Reality Assembly Guidance and Torque Monitoring System
- 110 -- Central Processing Unit
- 120 -- AR Head-Mounted Display (HMD) Subsystem

- 121 -- Forward-Facing RGB Camera
- 122 -- Inertial Measurement Unit (IMU)
- 123 -- Wireless Transceiver
- 130 -- CNN-Based Vision Subsystem
- 140 -- Assembly Guidance and Rendering Engine
- 141 -- Finite State Machine (FSM) Controller
- 150 -- Torque Monitoring Subsystem
- 151 -- Torque Sensor
- 152 -- Assembly Tool (Wrench / Nutrunner)
- 153 -- Torque Alert Engine
- 160 -- Proximity Sensor Subsystem
- 161 -- Ultrasonic Proximity Sensor
- 170 -- Assembly Traceability Logging Module
- 200 -- Assembly Sequence State S0 (Idle)
- 202 -- Assembly State S1: Part 2 inserted into Part 1 forming Sub-Assembly A1
- 204 -- Assembly State S2: Part 3 placed on A1 forming Sub-Assembly A2
- 206 -- Assembly State S3: Part 4 inserted through A2 forming Sub-Assembly A3
- 208 -- Assembly State S4: Part 5 locked into A3 forming Sub-Assembly A4
- 210 -- Assembly State S5: Torque verification and ultrasonic sensor confirmation
- 212 -- CNN Confidence Check (Threshold ≥ 0.85)
- 214 -- Torque Threshold Alert ($T_{\min} / T_{\max}$ comparison)
- 216 -- Ultrasonic Distance Verification ($d \leq 10$ cm)

DETAILED DESCRIPTION

Embodiments of the present disclosure will now be described with reference to the accompanying drawings.

Embodiments are provided so as to thoroughly and fully convey the scope of the present disclosure to the person skilled in the art. Numerous details are set forth, relating to specific components and methods, to provide a complete understanding of embodiments of the present disclosure. It will be apparent to the person skilled in the art that the details provided in the embodiments should not be construed to limit the scope of the present disclosure. In some embodiments, well-known processes, well-known apparatus structures, and well-known techniques are not described in detail.

In current industrial valve manufacturing, assembly lines rely predominantly on human operators following paper-based or static digital work instructions without real-time in-process verification, resulting in missing components, incorrect orientations, out-of-sequence assembly, and incorrect torque application. Existing computer vision inspection systems are post-assembly in nature and lack integration with AR guidance or torque monitoring workflows.

To address these limitations, an augmented reality based intelligent assembly guidance and torque monitoring system (hereinafter referred to as "system 100") and a corresponding method (hereinafter referred to as "method 200") are now described with reference to Figures 1 through 3.

Referring to Figure 1, the system 100 comprises a central processing unit 110 interconnecting all subsystems: an AR HMD subsystem 120, a CNN-based vision subsystem 130, an assembly guidance and rendering engine 140, a torque monitoring subsystem 150, a proximity sensor subsystem 160, and an assembly traceability logging module 170.

The AR HMD subsystem 120 is a binocular see-through display device incorporating a forward-facing RGB camera 121 with minimum resolution of 1920 x 1080 pixels at 30 frames per second, stereo display panels with a minimum 60-degree horizontal field-of-view, an inertial measurement unit 122, and a wireless transceiver 123. Suitable implementations include Microsoft HoloLens 2, Magic Leap 2, or equivalent industrial-grade AR HMD devices.

The CNN-based vision subsystem 130 employs a convolutional neural network architecture trained on a dataset of no fewer than 5,000 labelled images per component class. In the preferred embodiment, a YOLO v8 or ResNet-50 backbone is used for real-time inference. The CNN is configured to classify each valve sub-component, detect its bounding box, verify orientation alignment, and return a confidence score. Inference is performed on each camera frame received from camera 121.

The assembly guidance and rendering engine 140 implements a finite state machine 141 with states S0 (Idle) through S5 (Assembly Complete) as described in Figure 2. The five assembly states correspond to: S1 — inserting Part 2 (hollow knurled ring) into Part 1 (main body with handle) to form sub-assembly A1; S2 — placing Part 3 (end-cap disc) onto sub-assembly A1 to form A2; S3 — threading Part 4 (long needle pin) through A2 to form A3; S4 — inserting Part 5 (small dowel peg) through the cross-hole of Part 4 to lock and form A4; and S5 — final torque verification and ultrasonic sensor

confirmation. State transitions 212 are triggered exclusively by CNN inference confidence scores of 0.85 or greater, enforcing Poka-Yoke error-proofing at each step.

Upon valid state transition, the rendering engine 140 updates the AR holographic overlay in the operator's HMD field-of-view to display the next assembly step as a three-dimensional animation, with visual arrows indicating correct part placement positions and orientations. For each state, the operator may additionally upload an instructional video clip that plays within the AR display on the right-eye panel while the left-eye panel shows the live camera feed with CNN detection overlays.

The torque monitoring subsystem 150 comprises one or more torque sensors 151 embedded in or mechanically attached to assembly tools 152 such as wrenches and nutrunners. The sensors 151 communicate with the central processing unit 110 via Bluetooth 5.0 or wired RS-485 interface at a sampling rate of 100 Hz or greater. The torque alert engine 153 compares instantaneous torque value T against configurable thresholds T_{min} and T_{max} stored per fastener position in a component configuration database. When T is less than T_{min} , a yellow caution overlay is rendered in the AR HMD display. When T exceeds T_{max} , a red danger alert accompanied by an audible warning signal is generated, halting the current assembly step until safe torque parameters are restored.

The proximity sensor subsystem 160 uses an ultrasonic sensor 161, preferably an HC-SR04 or equivalent device operating at 40 kHz with measurement range of 2 centimeters to 400 centimeters and accuracy of plus or minus 3 millimeters. Final assembly completion is triggered at state S5 when the sensor 161 measures a distance d of 10 centimeters or less to the completed assembly unit presented by the operator, confirming the unit has been correctly assembled and presented to the verification station.

The assembly traceability logging module 170 records for each assembled valve unit: the unique unit serial number, operator identification, assembly start and completion timestamps, CNN confidence scores for each step, torque values measured at each fastener position, and the pass or fail verdict for each step. This log is stored locally and optionally synchronized to a cloud-based analytics module for long-term quality management and regulatory compliance.

In an embodiment, the system 100 comprises a simulation mode wherein the CNN-based detection is replaced by manual operator confirmation via a hardware button or on-screen control, enabling use of the guidance and torque monitoring system in environments where camera-based detection is temporarily unavailable.

In an embodiment, the system 100 supports multiple product configurations, enabling the assembly FSM, CNN model weights, torque threshold database, and 3D guidance content to be updated via a configuration management interface to accommodate different valve types and sizes without hardware modifications.

In an embodiment, the modular hardware and software architecture of system 100 supports integration with enterprise manufacturing execution systems (MES) and enterprise resource planning (ERP) systems via standard industrial communication protocols including OPC-UA and REST API interfaces.

TECHNICAL ADVANCEMENTS

The present disclosure described herein above has several technical advantages including, but not limited to, the realization of an augmented reality based intelligent assembly guidance and torque monitoring system and method that:

- eliminates assembly sequence errors through CNN-gated finite state machine control that prevents an operator from proceeding to any subsequent assembly step without verified correct completion of the current step, enforcing Poka-Yoke error-proofing at every stage;

- provides real-time in-process quality verification at each individual assembly step rather than post-assembly inspection, dramatically reducing rework costs, scrap rates, and production delays by detecting errors at the moment of occurrence;

- integrates real-time torque monitoring with configurable threshold-based AR alert generation, ensuring every fastener is tightened to the correct specification and eliminating under-torqued and over-torqued joints as sources of valve failure;

- accelerates operator training by providing immersive three-dimensional holographic assembly guidance, reducing the time required to train new operators compared to conventional paper-based or classroom instruction methods;

- generates comprehensive digital assembly traceability records for every assembled unit, including CNN confidence scores, torque values, and operator credentials, enabling full quality traceability required by ISO, ASME, and API manufacturing standards;

- provides a modular, adaptable architecture that supports multiple valve configurations, different CNN models, and various AR HMD platforms without requiring hardware modifications; and

- supports integration with existing enterprise manufacturing execution systems and cloud-based analytics platforms, enabling system-wide quality data aggregation and continuous improvement.

The embodiments herein and the various features and advantageous details thereof are explained with reference to the non-limiting embodiments in the following description. Descriptions of well-known components and processing techniques are

omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

The foregoing description of the specific embodiments so fully reveals the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation.

WE CLAIM:

CLAIM 1. An augmented reality based intelligent assembly guidance and torque monitoring system (100) for industrial valve manufacturing, the system (100) comprising:

- an AR head-mounted display (HMD) subsystem (120) comprising a forward-facing camera (121), stereo display panels, an inertial measurement unit (122), and a wireless transceiver (123), configured to overlay holographic assembly guidance content onto the operator's physical workspace;

- a CNN-based vision subsystem (130) comprising a convolutional neural network inference engine trained on labelled images of valve sub-components, configured to perform real-time part classification, orientation verification, and confidence score generation;

- an assembly guidance and rendering engine (140) implementing a finite state machine (141) with discrete states corresponding to each valve assembly step, wherein state transitions are triggered exclusively by CNN inference confidence scores meeting or exceeding a configurable threshold;

- a torque monitoring subsystem (150) comprising at least one torque sensor (151) communicatively coupled to an assembly tool (152) and a torque alert engine (153) configured to compare instantaneous torque values against pre-defined minimum and maximum thresholds per fastener position and generate visual alerts within the AR HMD upon threshold violation;

- a proximity sensor subsystem (160) comprising an ultrasonic sensor (161) configured to confirm final assembly completion when the measured distance falls below 10 centimeters; and

- an assembly traceability logging module (170) configured to record component identifiers, CNN confidence scores, torque measurements, timestamps, and operator credentials for each assembled unit,

wherein the assembly guidance and rendering engine (140) prevents the operator from proceeding to any subsequent assembly step without CNN-verified correct completion of the current step.

CLAIM 2. The system (100) as claimed in claim 1, wherein the CNN-based vision subsystem (130) employs a YOLO or ResNet-50 backbone architecture trained on a dataset comprising no fewer than 5,000 annotated images per valve component class, and wherein the configurable confidence threshold for state transition is 0.85.

CLAIM 3. The system (100) as claimed in claim 1, wherein the torque alert engine (153) is configured to render a yellow caution overlay in the AR HMD display when measured torque falls below the minimum threshold T_{min} , and a red danger alert with audible signal when measured torque exceeds the maximum threshold T_{max} , where T_{min} and T_{max} are configurable per fastener position stored in a component configuration database.

CLAIM 4. The system (100) as claimed in claim 1, wherein the assembly guidance and rendering engine (140) streams three-dimensional animation video content for each assembly step into the AR HMD stereo display, with the left-eye panel displaying the live camera feed with CNN detection bounding boxes and confidence scores, and the right-eye panel displaying the three-dimensional step-by-step assembly guidance animation.

CLAIM 5. The system (100) as claimed in claim 1, wherein the finite state machine (141) encodes five sequential assembly states corresponding to: insertion of a hollow ring component into a main body component forming a first sub-assembly; placement of an end-cap disc onto the first sub-assembly forming a second sub-assembly; threading of a needle pin through the second sub-assembly forming a third sub-assembly; locking of a dowel peg into the needle pin forming a fourth sub-assembly; and torque verification and ultrasonic sensor confirmation of the completed assembly.

CLAIM 6. The system (100) as claimed in claim 1, further comprising a cloud-based analytics module configured to store assembly traceability logs, CNN performance metrics, torque records, and system usage data to support long-term quality management and regulatory compliance.

CLAIM 7. The system (100) as claimed in claim 1, wherein the torque sensor (151) communicates with the central processing unit (110) via Bluetooth 5.0 or wired RS-485 interface at a sampling rate of 100 Hz or greater.

CLAIM 8. The system (100) as claimed in claim 1, further comprising a simulation mode wherein CNN-based detection is replaced by manual operator confirmation, enabling operation of the guidance and torque monitoring functions in environments where camera-based detection is temporarily unavailable.

CLAIM 9. The system (100) as claimed in claim 1, wherein the system (100) supports multiple product configurations enabling the finite state machine, CNN model weights, torque threshold database, and three-dimensional guidance content to be updated via a configuration management interface to accommodate different valve types and sizes without hardware modifications.

CLAIM 10. A method for augmented reality guided industrial assembly with real-time torque monitoring, the method comprising:

detecting and classifying individual valve sub-components using a trained CNN model receiving image frames from a camera (121) integrated into an AR head-mounted display (HMD) subsystem (120);

displaying three-dimensional holographic assembly guidance overlays within the operator's field-of-view via the AR HMD upon successful component detection and orientation verification by the CNN model;

gating each assembly state transition using the finite state machine (141) such that the operator cannot proceed to a subsequent step unless the CNN inference confidence score meets or exceeds a threshold of 0.85;

monitoring real-time torque applied to fasteners via torque sensors (151) integrated with or attached to assembly tools (152) during each fastener tightening operation;

comparing measured torque values against stored minimum and maximum threshold values per fastener position and generating AR-rendered visual and audible alerts within the HMD display upon threshold violation;

confirming final assembly completion via an ultrasonic sensor (161) when the measured distance between the sensor and the completed assembly is 10 centimeters or less; and

logging all assembly step data including component identifiers, CNN confidence scores, torque values per joint, timestamps, and operator credentials in an assembly traceability logging module (170) for quality records.

Dated this 23 day of March, 2025

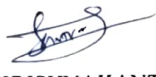

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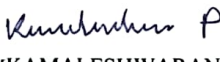

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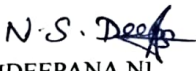

[SANGEETHA S]


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[KRISHNAKANTH J]


[KAMALESHWARAN P]


[TITUS PAUL H]


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TO,
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THE PATENT OFFICE, AT CHENNAI

ABSTRACT

AUGMENTED REALITY BASED INTELLIGENT ASSEMBLY GUIDANCE AND TORQUE MONITORING SYSTEM FOR INDUSTRIAL VALVE MANUFACTURING

An augmented reality based intelligent assembly guidance and torque monitoring system (100) for industrial valve manufacturing is disclosed. The system (100) comprises an AR head-mounted display (HMD) subsystem (120) with an integrated forward-facing camera (121), a CNN-based vision subsystem (130) trained on labelled images of valve sub-components, an assembly guidance and rendering engine (140) implementing a finite state machine (141), a torque monitoring subsystem (150) comprising torque sensors (151) coupled to assembly tools (152) and a torque alert engine (153), a proximity sensor subsystem (160) comprising an ultrasonic sensor (161), and an assembly traceability logging module (170). The CNN module (130) performs real-time part classification and orientation verification; state transitions are gated by CNN confidence scores of 0.85 or greater, enforcing Poka-Yoke error-proofing at each step. The torque alert engine (153) compares real-time torque values against configurable thresholds per fastener position and renders yellow caution or red danger overlays within the AR HMD upon violation. Final assembly completion is confirmed by the ultrasonic sensor (161) when the measured distance is 10 centimeters or less. All assembly data is logged in the traceability module (170) for full quality traceability. The system (100) substantially eliminates assembly errors, enforces torque compliance, accelerates operator training, and enables full digital traceability in industrial multi-component valve manufacturing.